Name: ______________________

Worksheet

Justifying terms of a sequence

Task A: Justifying terms of a sequence

1) Will 467 be in the arithmetic sequence starting $-6, 2, 10, 18, \dots$? Justify your answer.

2) 113 and 129 are consecutive terms in an arithmetic sequence. Will $12\,718$ be in the sequence? Justify your answer.

3) Will 467 be in the arithmetic sequence starting $-6, -1, 4, \dots$? Justify your answer.

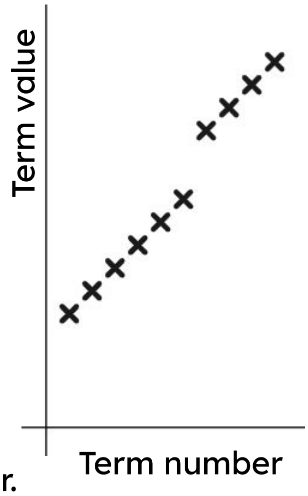
4) 113 and 123 are consecutive terms in an arithmetic sequence. Will $12\,713$ be in the sequence? Justify your answer.

Task B: Using a graphical representation

1) Sam is asked to continue this arithmetic sequence and plot it.

127 850, 135 350, 142 850, …

They write the below values and plot them.

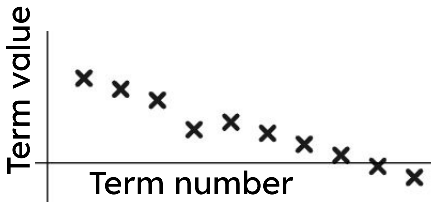


127 850, 135 350, 142 850, 150 350, 157 850,
165 350, 178 250, 185 750, 193 250, 200 750, …

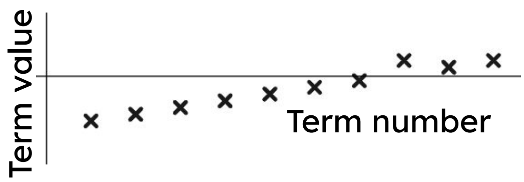
Write a sentence explaining how you know there has been an error.

2) Use the graphs to spot the incorrect terms in these arithmetic sequences.

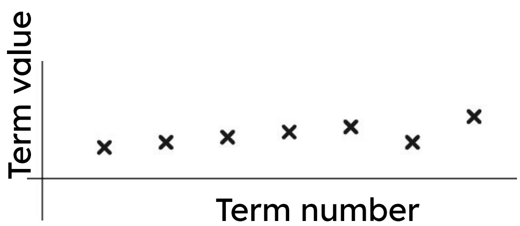
a) $0.83, 0.72, 0.61, 0.3, 0.39, 0.28, 0.17, 0.06, -0.05, -0.16, \ldots$



b) $-0.1, -0.85, -0.7, -0.55, -0.4, -0.25, -0.1, 0.35, 0.2, 0.35, \ldots$

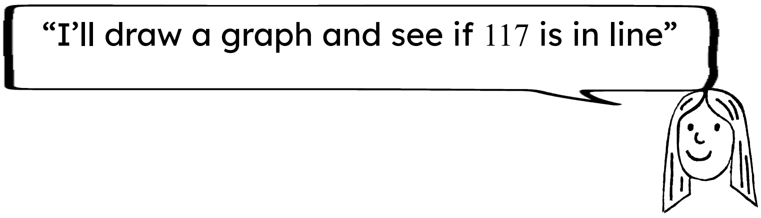


c) $\frac{1}{2}, \frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}, \frac{9}{15}, 1 \ldots$



3) Laura is asked if the sequence 117 is in the sequence starting $-8, -2, 4, 10, 16, \ldots$

She says:



Write a sentence explaining why this might not be the most efficient method to use on this occasion.

Task C: Using the n^{th} term

1) Use n^{th} term and approximation to find 400 in the arithmetic sequence starting $10, 23, 36, 49, \ldots$

2) Is 400 in the arithmetic sequence starting $-22, -14, -6, 2, \ldots$

Write a sentence to justify your answer.

3) Is -12.75 in the arithmetic sequence starting $21.25, 20.9, 20.55, 20.2, \ldots$

Fully justify your answer.